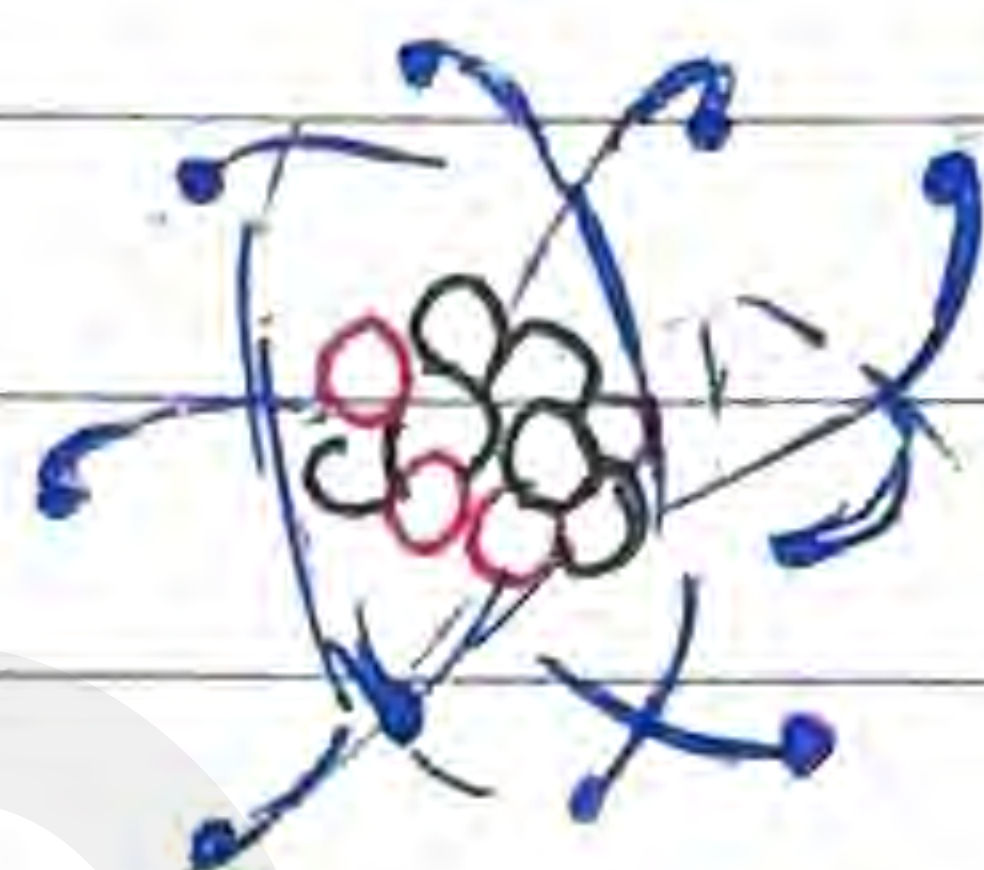


Nuclear Physics



e^- electrons

p^+ protons

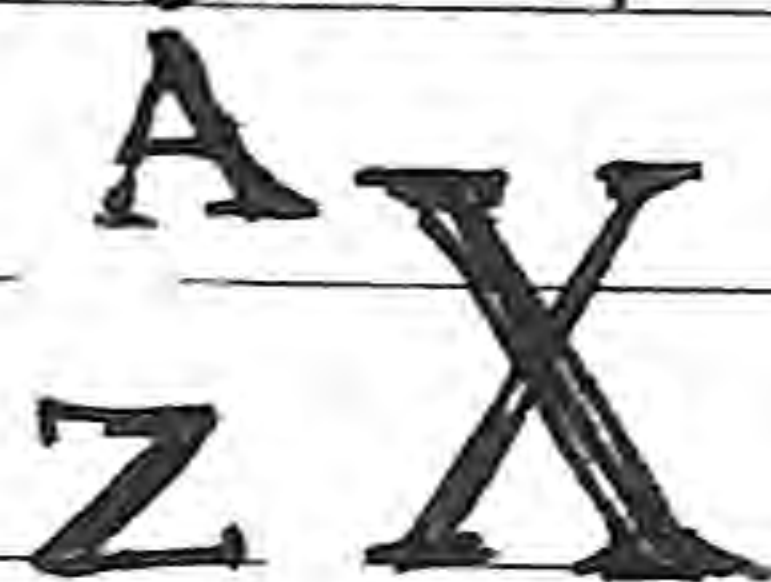
n^0 neutrons

$$m_{p^+} \approx m_{n^0} \approx 1800 m_{e^-}$$

↙ negligible mass

Charges are found in the electron cloud (-) or nucleus (+)

Notation



Z - Atomic #, # of protons in the nucleus

N - Neutron #, # of neutrons in the nucleus

A - Atomic mass #, # of nucleons in the nucleus

Nucleus consists of neutrons and protons
neutrons = $n = A - Z$

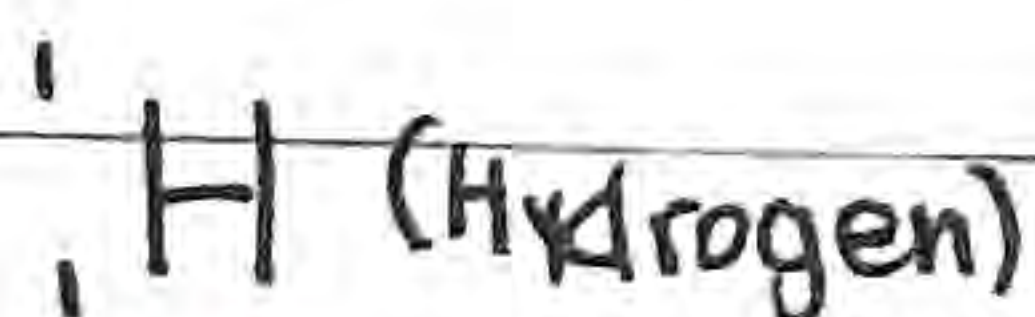
Nucleus (centre of atom)

• Nuclei (multiple nucleus)

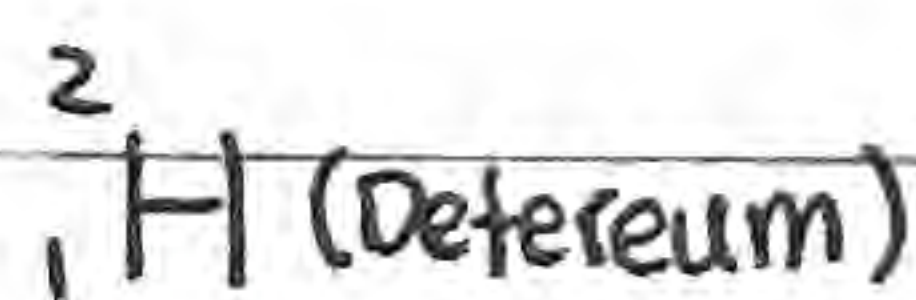
• Nuclide (distinct type of nucleus)

Isotopes - Same element but of different # of neutrons

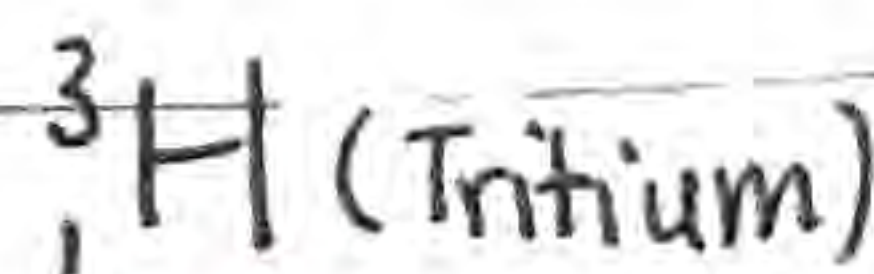
Hydrogen...



$1p^+$
 $0n^0$



$1p^+$
 $1n^0$



$1p^+$
 $2n^0$

Nuclear Physics

Evidence for neutrons

- Isotopes - No other way to explain mass difference of two isotopes of the same element
 - ↳ Same element = same # of p^+ = same p^+ mass
 - ↳ As such, must be some other particle as well (n^0)

Summary of Masses

Particle	kg	u	MeV/c ²
Proton	1.6726×10^{-27}	1.007276	938.28
Neutron	1.6750×10^{-27}	1.008665	939.57
Electron	9.109×10^{-31}	5.486×10^{-4}	0.511
¹ H atom	1.6735×10^{-27}	1.007825	938.78

atomic mass unit (use all s.f. bc of imprecision)

MeV/c² is from ...

$$eV = \text{Energy} = mc^2$$

$$\text{Thus } m = \frac{eV}{c^2} \text{ with prefix M} = 10^6$$

atomic mass unit is also g·mol⁻¹
(periodic table given)

Binding Energy

In the macro world, combining two objects together yields a mass equal to the sum of the two smaller objects

↳ But in the nuclear world, every tiny bit of mass matters

↳ When you combine two objects, they release energy

(making bonds = exothermic) $\Rightarrow$ **Nucleus**

that's why

H has no binding energy

↳ To release energy, mass is converted (the mass converted into energy is known as mass defect)

$$E = mc^2 \quad \leftarrow \text{conversion mass to energy}$$

In the nuclear world, every energy released matters

Since $m = \frac{E}{c^2}$ is of similar magnitude to other masses. $\rightarrow$ in the macro world, masses are already big so Δm negligible

Thus... separate = together + energy released

* Protons + Neutrons > Mass Nucleus

$$\text{Mass Defect} = \Delta \text{mass} = (\text{Protons} + \text{Neutrons}) - \text{Nucleus}$$



$$\text{Energy Released} = \Delta m \cdot c^2$$

if asking for MeV for energy then...

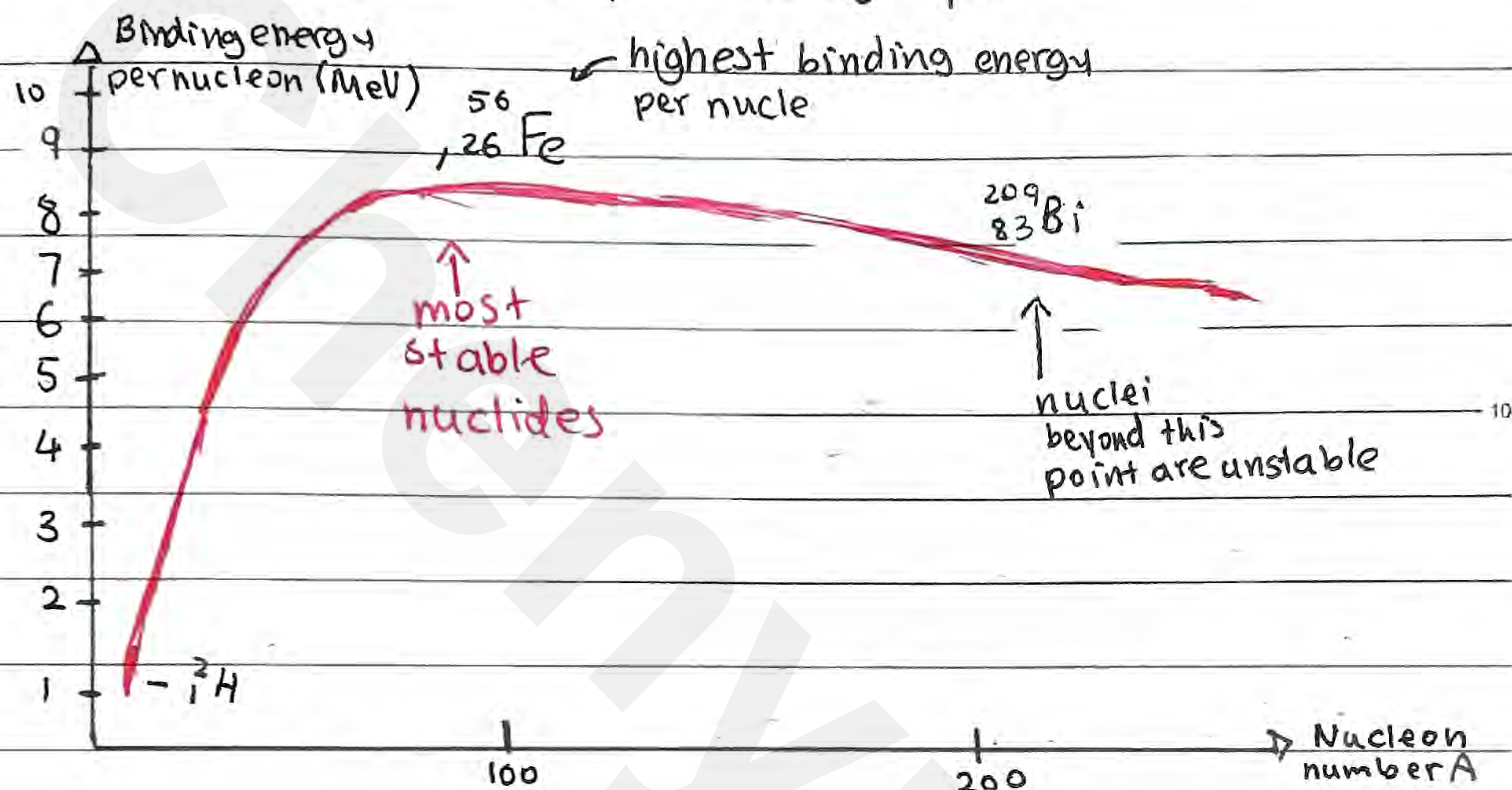
$$\frac{BE}{c^2} = \Delta m$$

use conversion from u to $\frac{\text{MeV}}{c^2}$

↳ typically not using Joules in nuclear physics thus... $BE = \text{--- MeV}$

Binding Energy

When you calculate the binding energy per nucleon for successive elements, you can graph it...



Since ${}^{56}_{26}\text{Fe}$ is the one requiring most binding energy per nucleon, we know it'll be formed in the center of stars:



Stability

- Nuclear stability depends on # of protons with neutrons
- This stability deviates from 1:1 ratio

(requires more neutrons per proton as # of protons increase)

↳ Since greater repulsive electromagnetic force (act on range = ∞)

↳ An attractive force is required to overcome it

- Strong Nuclear Force (range = nucleus $\approx 10^{-10}\text{m}$; only acts on nucleons)

Nuclear Stability



Stability depends on potential energy

- Low potential energy = stable (ground state)
- High potential energy = unstable (excited state)